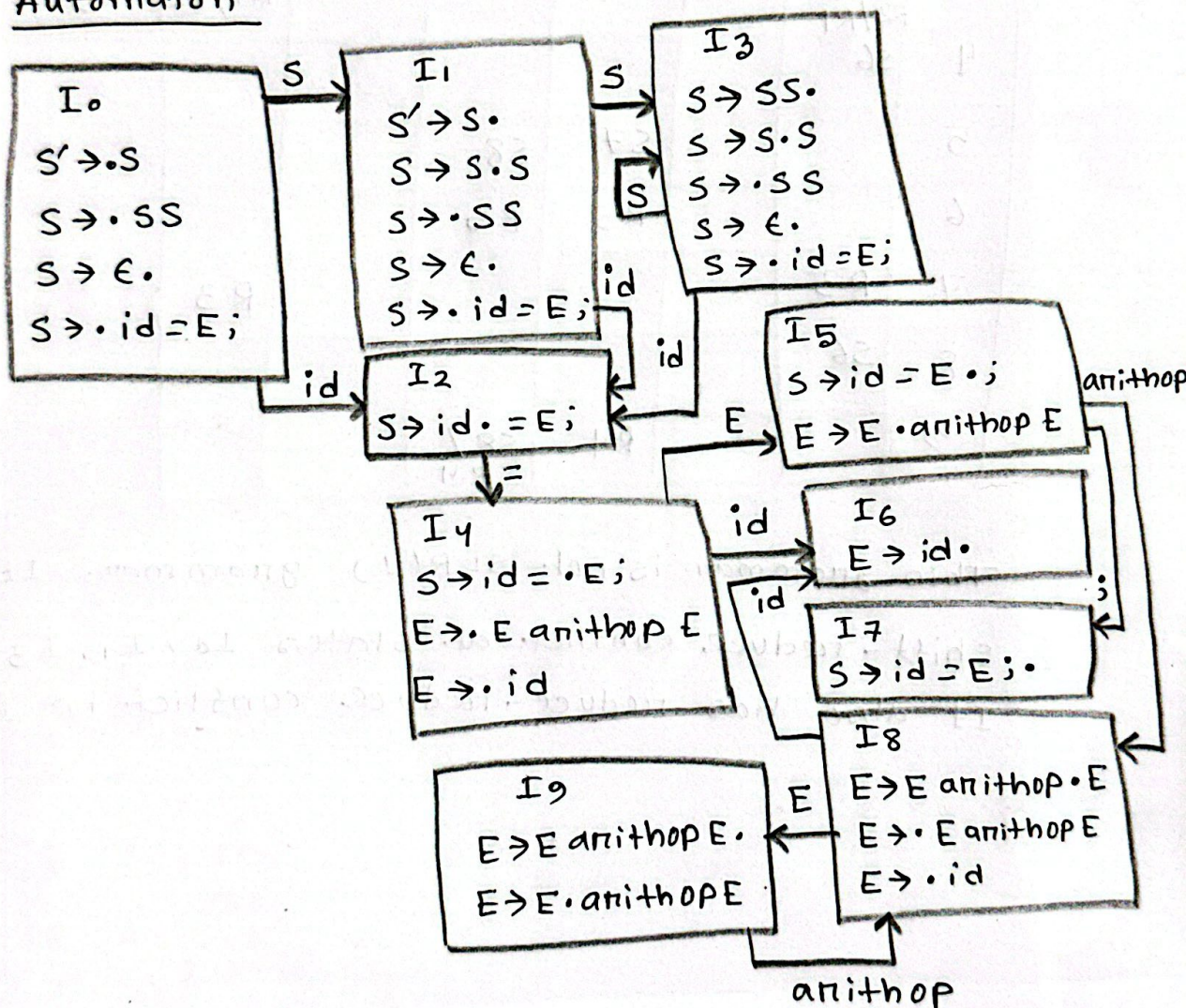


Spring 2026 (A)

③ a) Augment : $S' \rightarrow S$ 1. $S \rightarrow SS$ 2. $S \rightarrow \epsilon$ 3. $S \rightarrow id = E ;$ 4. $E \rightarrow E \text{ anithop } E$ 5. $E \rightarrow id$ Automaton

SLR Parsing Table:

State	Action						Go To	
	id	=	;	anithop	€	\$	S	E
0	S2/ R2					R2	1	
1	S2/ R2					Accept	3	
2		S4						
3	S2/ R2/R1					R1/R2	3	
4	S6							5
5			S7	S8				
6			R5	R5				
7	R3					R3		
8	S6							9
9			R4	S8/ R4				

This grammar is not SLR(1) grammar. It has shift-reduce conflict at states I_0 , I_1 , I_3 and I_9 . It also has reduce-reduce conflict in I_3 .

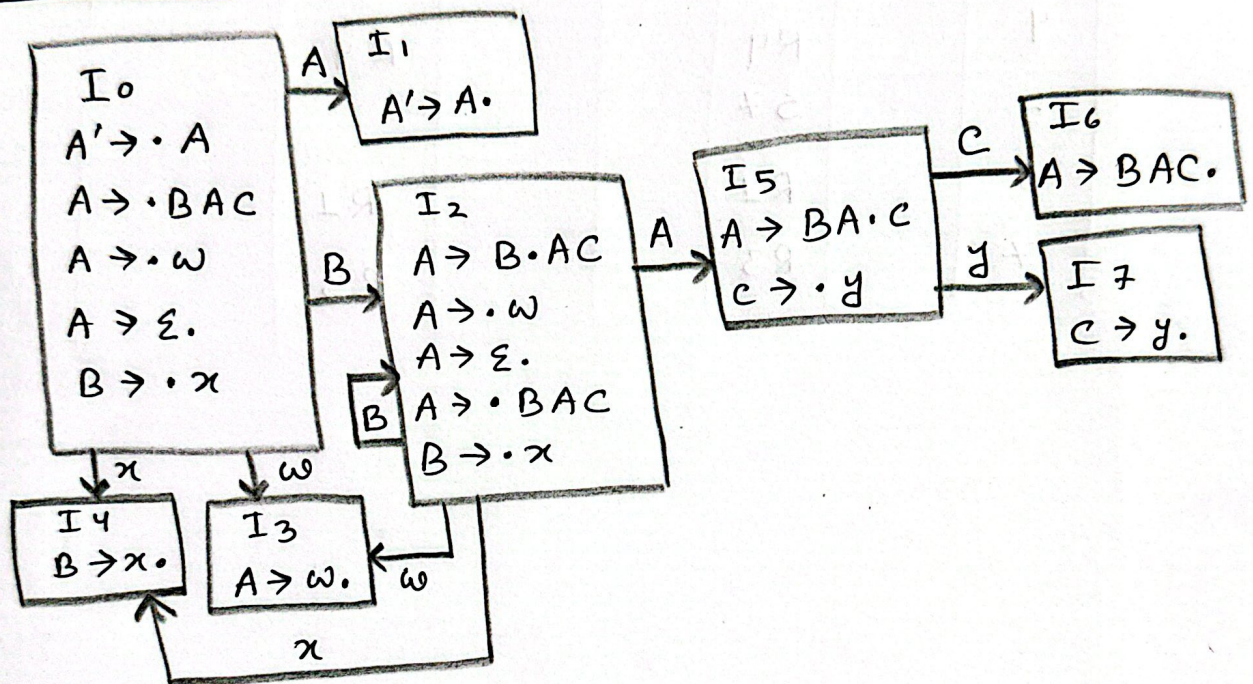
Summer 2025 (A)

③ SLR(1) Parse Table

State	Action						Go To			
	while	if	(	a	)	\$	S	stmt	Block	Cond
0	S4	S3		S5		R2	1	2		
1						Accept				
2	S4	S3		S5		R2	6	2		
3			S7							
4			S8							
5	R5	R5		R5		R5				
6						R1				
7				S11						9
8				S11						10
9					S12					
10					S13					
11					R7					
12	R6	R6		R6		R6			14	
13	R6	R6		R6		R6			15	
14	R3	R3		R3		R3				
15	R4	R4		R4		R4				

Stack	Symbol	Input	Action
0	\$	if(a) while(a) a \$	shift to 3
03	\$ if	(a) while(a) a \$	shift to 7
037	\$ if (	a) while(a) a \$	shift to 11
03711	\$ if (a	) while(a) a \$	reduce by cond $\rightarrow a$
0379	\$ if (cond	) while(a) a \$	shift to 12
037912	\$ if (cond)	while(a) a \$	reduce by Block $\rightarrow \epsilon$
03791214	\$ if (cond) Block	while(a) a \$	reduce by stmt $\rightarrow if$ (cond) Block
02	\$ stmt	while(a) a \$	shift to 4
024	\$ stmt while	(a) a \$	shift to 8
0248	\$ stmt while (	a) a \$	shift to 11
024811	\$ stmt while (a	) a \$	reduce by cond $\rightarrow a$
024810	\$ stmt while (cond	) a \$	shift to 13
02481013	\$ stmt while (cond)	a \$	reduce by Block $\rightarrow \epsilon$
0248101315	\$ stmt while (cond) Block	a \$	reduce by stmt $\rightarrow while$ (cond) Block
022	\$ stmt stmt	a \$	shift to 5
0225	\$ stmt stmt a	\$	reduce by stmt $\rightarrow a$
0222	\$ stmt stmt stmt	\$	reduce by S $\rightarrow \epsilon$
02226	\$ stmt stmt stmt S	\$	reduce by S $\rightarrow stmt S$
0225	\$ stmt stmt S	\$	reduce by S $\rightarrow stmt S$
026	\$ stmts	\$	reduce by S $\rightarrow stmts$
01	\$ S	\$	Accept

Spring 2025 (A)

② Augmented $A' \rightarrow A$ 1. $A \rightarrow BAC$ 2. $B \rightarrow x$ 3. $C \rightarrow y$ 4. $A \rightarrow w$ 5. $A \rightarrow \epsilon$ Automaton $\text{First}(A) = \{x, w, \epsilon\}$ $\text{First}(B) = \{x\}$ $\text{First}(C) = \{y\}$ $\text{Follow}(A) = \{y, \$\}$ $\text{Follow}(B) = \{x, w, y, \$\}$ $\text{Follow}(C) = \{y, \$\}$

SLR(1) Parse Table

State	Action					GoTo		
	x	y	w	e	\$	A	B	c
0	S4	R5	S3		R5	1	2	
1					Accept			
2	S4	R5	S3		R5	5	2	
3		R4			R4			
4	R2	R2	R2		R2			
5		S7						6
6		R1			R1			
7		R3			R3			

④

State	Symbol	Input	Action
0	\$	((id+(id)))\$	shift to 4
04	\$(	(id+(id)))\$	shift to 4
044	\$((	id+(id)))\$	shift to 5
0445	\$((id	+(id)))\$	reduce by $F \rightarrow id$
0443	\$((F	+(id)))\$	reduce by $T \rightarrow F$
0442	\$((T	+(id)))\$	reduce by $E \rightarrow T$
0448	\$((E	+(id)))\$	shift to 6
04486	\$((E+	(id)))\$	shift to 4
044864	\$((E+(	id)))\$	shift to 5
0448645	\$((E+(id	)))\$	reduce by $F \rightarrow id$
0448643	\$((E+(F	)))\$	reduce by $T \rightarrow F$
0448642	\$((E+(T	)))\$	reduce by $E \rightarrow T$
0448648	\$((E+(E	)))\$	shift to 11
044864811	\$((E+(E)	)))\$	reduce by $F \rightarrow (E)$
044863	\$((E+F	)))\$	reduce by $T \rightarrow F$
044869	\$((E+T	)))\$	reduce by $E \rightarrow E+T$
0448	\$((E	)))\$	shift to 11
044811	\$((E)	))\$	reduce by $F \rightarrow (E)$
043	\$(F	)\$	reduce by $T \rightarrow F$
042	\$(T	)\$	reduce by $E \rightarrow T$
048	\$(E	)\$	shift to 11
04811	\$(E)	\$	reduce by $F \rightarrow (E)$
03	\$F	\$	reduce by $T \rightarrow F$
02	\$T	\$	reduce by $E \rightarrow T$
01	\$E	\$	Accept

Fall 2025(A)

③

Stack	Symbol	Input	Action
0	\$	while id do end\$	shift to 5
05	\$ while	id do end \$	shift to 7
057	\$ while id	do end \$	reduce by $E \rightarrow id$
058	\$ while E	do end \$	shift to 9
0589	\$ while E do	end \$	reduce by $S \rightarrow E$
058910	\$ while E do S	end \$	shift to 11
05891011	\$ while E do S end	\$	reduce by $w \rightarrow$ while E do S end
03	\$ w	\$	reduce by $A \rightarrow w$
02	\$ A	\$	reduce by $S \rightarrow E$
026	\$ A S	\$	reduce by $S \rightarrow AS$
01	\$ S	\$	Accept